



CHAPTER 5 SIMPLE TESTING

PHOTO: JOSH LARSON

Testing and assessment are really coming on in the climbing training world, and for good reason. For years, we've looked at maximum redpoint grades, onsights, and competition placements as markers, but these performance numbers are the sum of much more than acute improvements in training variables. The goal of coaches and sport scientists looking at testing for climbing is to build reliable, accurate, and simple tests to let athletes know whether their strength and conditioning is improving. We want to be able to test at the beginning of training and at the end of it, and see an improvement along the way.

The hard part of designing testing is to find a measure that accurately predicts improved performance on the rock. Because of the complexity of the sport, this is very difficult to do - your fingers can indeed get stronger but it does nothing to reduce the paralyzing fear you feel (your true limiter) when facing a short fall from above a bolt. With this in mind, I strongly suggest that you employ the assistance of a trusted climbing partner or a coach, as well as video. By asking about what your friend sees and by actually watching yourself climb, you can gain all kinds of insight into performance that goes beyond your physical training.

To test for specific strength and conditioning variables, I suggest you employ even very simple tests occasionally throughout your training. My personal favorite method is to run a full testing series (usually about 30 minutes) once per training phase.

In general, you should be looking for slight improvements in the facet of training that you have just been focusing on, but the thing

I am most interested in is making sure that the numbers you have not been focusing on don't drop too far when you're *not* focusing on them.

With this in mind, you can select any of a number of tests aimed at specific measures of fitness. The important part of testing is consistency: you want your testing equipment and environment to be the same each time to try and control for as many variables as possible.

A quick aside: Our gym has several barbells. We have several standard 20kg / 44 pound bars, some specialty bars, and we have two women's competition bars that clock in at 15kg (33 pounds). Although they are clearly smaller when placed next to a standard bar, the women's bars don't look all that different. Once every few months, I'll come into the gym in the morning and find one of these smaller bars racked in the bench press. It's really unlikely a female competitive lifter picked this bar for the bench - and more likely that someone accidentally picked that one, did their bench press workout, and maybe set an all new "personal best" without knowing the bar they were using was 11 pounds light. Consistency matters.

When setting up for testing, make sure you use the same equipment, same testing tools, use the same chalk / shoes / tape, and test at the same time of the month. You should do the same warm-up each testing day, and should aim to do the tests in the same order.

FINGER STRENGTH TESTING

We can go pretty deep here, and there is a lot of interest in this testing these days. I suggest you do two finger strength measures.

1. 10-second maximum-load 2-hand hang on a medium (14-18mm) edge. Weigh yourself on testing day, and hang with a load for 10 seconds. Rest a lot, then do it again with 5

more pounds. Micro-adjusting isn't necessary - if you can't go up by 5 in a month, you didn't get that much stronger.



2. Single-arm force production on a 20mm edge. Get a Tension Block and a crane scale (or Tindeg), and set them up on a rack or rafter so that you can pull. Set up so the elbow is about 120 degrees, and then test a couple of pulls each side. Stronger climbers will have to brace their bodies to avoid pulling up off the ground. If you can consistently do more than bodyweight, consider moving to a 15mm edge for testing.



Both of these tests are easy to set up and are quick. Doing them once a month in your training cycles is informative (especially to keep you from freaking out and thinking you are getting weaker in non-strength phases) and useful for programming upcoming training.

UPPER BODY STRENGTH TESTING

I like the 3-rep max pull-up here. We can cheat singles with shallow negatives and neck stretching at the top, but triples don't lie. Straight elbows at the bottom, entire head above the bar and eyes still facing the wall in front of you. Respect the exercise. Like the finger testing above, 5 pound or 2kg increments are good.



TOTAL BODY STRENGTH TESTING

The deadlift is a great test here. If you don't know how to do them, hire a strength coach to teach you. If they hurt your back, refer to the last sentence. Trap bar deadlifts are probably better for most people. Do a 3-rep max here, too. You won't lift as much as in a single, but the test is to show progress in total strength, not achievement in one exercise.



TOTAL BODY POWER TESTING

Vertical jump is a simple and very informative test. Get the Jump Power app or put a tape measure on your wall. Jump as high as you can. Record it. If you are gaining power, this number will increase over time. Another interesting use of the vertical jump is to test nervous system fatigue. If you want to see if you are having problems recovering from particularly heavy or powerful training, vertical jump will tell you. Once you have a good baseline set, you can test vertical jump during a session's warm-up. If the numbers are down (by 15% or more), you're still not recovered and should opt for less-intense training that day.



UPPER BODY POWER TESTING

For this, we use the simple 10 second push-up test. It is a perfect test for climbers because it doesn't overemphasize their strengths, and gives clear numbers on power production. Using PERFECT form, do as many reps as you can in just ten seconds. Rest 3-5 minutes, then repeat the test a second time. Rest again for 3-5 minutes, and do a third set. Take the average of the three.

You can also do a 10 second pull-up test. I recommend this only as an additional test to the push-ups, not a replacement for them. Follow the same protocol, and record your average of 3.



LOCAL MUSCULAR ENDURANCE TESTING

The muscles of the forearm are really the only ones that need tremendous levels of muscular endurance for hard rock climbing. Even on endurance routes, your upper body and core and legs get plenty of rest, and are used primarily for generating strength and power. Therefore, most climbers only really have to test the forearms. I like two different tests for this.

1. Foot-on edge ladders. Do these on a campus board with foot rails or a little stool placed beneath the board. Set a timer and count the reps as you ladder up and down the rungs, alternating left and right. Don't move the feet up to higher foot rails or holds. Count the reps, and go until you fall off. Record both time and reps. Don't worry about skipping rungs - it can get complicated in a hurry and your testing numbers might be less consistent.



PHOTO: MEI RATZ

2. Large edge repeats on hangboard. This one is a bit boring, but is good if you only have a hangboard. Pick an 18-20mm edge (preferably on a wood board), and set an interval timer to repeat a 10-second on, 10-second off clock. Start by

hanging 10, then step down, drop your arms to the sides and rest 10. Repeat until you can't hang 10. Record the number of successful hangs.

You don't need to do both tests - they should tell a similar story. Pick one and do it *after* any strength testing, once per cycle.



GENERAL ENDURANCE TESTING

General endurance testing is really common in sports science. We aren't looking for a test of aerobic capacity or top end speed, we simply want to see if you can put out more work in a limited amount of time. For this reason, I like a 10 minute test on a rowing machine or air bike. Warm-up, then go 10 minutes at a hard steady pace and record your total output - calories are the most tangible measure. Watch for pushing too hard over the anaerobic threshold early in the test - your power will drop off and you'll really suffer for the rest of the 10 minutes. Over the course of a general endurance phase, you should see this number go up steadily. It should be easy to maintain through the specific endurance phase, and should drop only slightly in strength and power phases. If it drops too much, your maintenance conditioning volume is too low.



Remember that the tests are simply measures of very specific facets of your fitness. They won't tell you how your skills are progressing, and they won't tell you how well you are integrating all of your fitness factors together.

Don't get lost in assessment. Too often, I see climbers modifying everything about their training in order to perform well in tests, and losing focus on getting better on the rock.